

# Operational Definition of the Phoenix Sepsis Score

July 23, 2026

## Table of contents

|          |                                                 |          |
|----------|-------------------------------------------------|----------|
| <b>1</b> | <b>Objective</b>                                | <b>3</b> |
| <b>2</b> | <b>Notation</b>                                 | <b>3</b> |
| 2.1      | Symbols . . . . .                               | 3        |
| 2.2      | Function Definitions . . . . .                  | 5        |
| <b>3</b> | <b>Defining Organ System Dysfunction Scores</b> | <b>6</b> |
| 3.1      | Respiratory . . . . .                           | 6        |
| 3.1.1    | Data from the EHR . . . . .                     | 6        |
| 3.1.2    | Constructed Variables . . . . .                 | 7        |
| 3.1.3    | Respiratory Organ Dysfunction Score . . . . .   | 8        |
| 3.2      | Cardiovascular . . . . .                        | 8        |
| 3.2.1    | Data from the EHR . . . . .                     | 9        |
| 3.2.2    | Constructed Variable . . . . .                  | 9        |
| 3.2.3    | Cardiovascular Dysfunction Score . . . . .      | 11       |
| 3.3      | Neurological . . . . .                          | 11       |
| 3.3.1    | Data from the EHR . . . . .                     | 11       |
| 3.3.2    | Constructed Variable . . . . .                  | 11       |
| 3.3.3    | Neurological Organ Dysfunction Score . . . . .  | 12       |
| 3.4      | Coagulation . . . . .                           | 12       |
| 3.4.1    | Data from the EHR . . . . .                     | 12       |
| 3.4.2    | Constructed Variable . . . . .                  | 13       |
| 3.4.3    | Coagulation Organ Dysfunction Score . . . . .   | 13       |
| 3.5      | Endocrine . . . . .                             | 13       |
| 3.5.1    | Data from the EHR . . . . .                     | 13       |
| 3.5.2    | Constructed Variables . . . . .                 | 13       |
| 3.5.3    | Endocrine Organ Dysfunction Score . . . . .     | 13       |
| 3.6      | Immunologic . . . . .                           | 14       |
| 3.6.1    | Data from the EHR . . . . .                     | 14       |
| 3.6.2    | Constructed Variables . . . . .                 | 14       |
| 3.6.3    | Immunologic Organ Dysfunction Score . . . . .   | 14       |
| 3.7      | Hepatic . . . . .                               | 14       |
| 3.7.1    | Data from the EHR . . . . .                     | 14       |
| 3.7.2    | Constructed Variables . . . . .                 | 14       |
| 3.7.3    | Hepatic Organ Dysfunction Score . . . . .       | 15       |
| 3.8      | Renal . . . . .                                 | 15       |
| 3.8.1    | Data from the EHR . . . . .                     | 15       |
| 3.8.2    | Constructed Variables . . . . .                 | 15       |
| 3.8.3    | Renal Organ Dysfunction Score . . . . .         | 15       |

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| 3.9 Suspected Infection . . . . .                               | 16        |
| <b>4 Phoenix Sepsis Score</b>                                   | <b>16</b> |
| <b>5 Exploratory Aggregation Schemes</b>                        | <b>17</b> |
| 5.1 Organ-Level Maxima . . . . .                                | 17        |
| 5.2 Organ-Level + Cardiovascular Component Decoupling . . . . . | 18        |
| 5.3 Full Component Decoupling . . . . .                         | 18        |

## List of Tables

|                                                                                                                                       |    |
|---------------------------------------------------------------------------------------------------------------------------------------|----|
| 1 Notation used in the operational definition of the Phoenix Sepsis Score. . . . .                                                    | 3  |
| 2 Logic with NA. . . . .                                                                                                              | 5  |
| 3 EHR Inputs for Respiratory Organ Dysfunction Scoring . . . . .                                                                      | 6  |
| 4 EHR Inputs for Cardiovascular Organ Dysfunction Scoring . . . . .                                                                   | 9  |
| 5 EHR Inputs for Neurological Organ Dysfunction Scoring . . . . .                                                                     | 11 |
| 6 EHR Inputs for Coagulation Organ Dysfunction Scoring . . . . .                                                                      | 13 |
| 7 EHR Inputs for Endocrine Organ Dysfunction Scoring . . . . .                                                                        | 13 |
| 8 EHR Inputs for Immunologic Organ Dysfunction Scoring . . . . .                                                                      | 14 |
| 9 EHR Inputs for Hepatic Organ Dysfunction Scoring . . . . .                                                                          | 14 |
| 10 EHR Inputs for Renal Organ Dysfunction Scoring . . . . .                                                                           | 15 |
| 11 Parameter values defining time windows, look-backs, and thresholds for the Phoenix Sepsis Score, Sepsis, and Septic Shock. . . . . | 17 |

# 1 Objective

Provide formal notation for the operational definition of the Phoenix Sepsis Score as published in [Sanchez-Pinto et al. \[2024\]](#) and [Schlapbach et al. \[2024\]](#) along with several exploratory aggregation schemes. The different scoring methods are based on when and how data is aggregated. The differences in the aggregation methods can dramatically change the Phoenix score and the published cut point of two or more points for sepsis may not be applicable to all aggregations.

## 2 Notation

### 2.1 Symbols

This section is a quick guide to the symbols used throughout the document. Detailed definitions for the functions will follow in § 2.2

Table 1: Notation used in the operational definition of the Phoenix Sepsis Score.

| Symbol                                     | Meaning                                                                                                 | Example or Notes                                                                                                                                                    |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Sets and Membership</b>                 |                                                                                                         |                                                                                                                                                                     |
| $\{\cdot\}$                                | Set notation used to define collections of values satisfying a condition.                               | $\{x : x \leq t\}$ denotes the set of values $x$ less than or equal to $t$ . Also, $A = \{1, 3, 5\}$ is read as "the set A consisting of the elements 1, 3, and 5." |
| $\emptyset$                                | Empty set.                                                                                              | Represents a set containing no elements.                                                                                                                            |
| $\in$                                      | Set membership operator.                                                                                | $t \in T$ indicates that time $t$ lies within the time window $T$ .                                                                                                 |
| $A \setminus B$                            | Remove elements of a set                                                                                | Example $\{1, 2, 3\} \setminus \{2\} = \{1, 3\}$                                                                                                                    |
| <b>Intervals</b>                           |                                                                                                         |                                                                                                                                                                     |
| $(a, b)$                                   | Open interval.                                                                                          | $X \in (a, b) \equiv a < X < b$ .                                                                                                                                   |
| $[a, b)$                                   | Left-closed, right-open interval.                                                                       | $X \in [a, b) \equiv a \leq X < b$ .                                                                                                                                |
| $(a, b]$                                   | Left-open, right-closed interval.                                                                       | $X \in (a, b] \equiv a < X \leq b$ .                                                                                                                                |
| $[a, b]$                                   | Closed interval.                                                                                        | $X \in [a, b] \equiv a \leq X \leq b$ .                                                                                                                             |
| <b>Definitions, Equality, and Logic</b>    |                                                                                                         |                                                                                                                                                                     |
| $:=$                                       | Definition operator indicating that the quantity on the left is defined by the expression on the right. | $f(x) := x^2$ defines the function $f(x)$ as $x^2$ .                                                                                                                |
| $=$                                        | Equality relation.                                                                                      | $2 + 3 = 5$ .                                                                                                                                                       |
| $\equiv$                                   | Logical equivalence or identity.                                                                        | $A \equiv B$ means that $A$ and $B$ are equivalent statements.                                                                                                      |
| $\vee$                                     | Logical OR operator.                                                                                    | $A \vee B$ is true if either $A$ or $B$ is true.                                                                                                                    |
| $\wedge$                                   | Logical AND operator.                                                                                   | $A \wedge B$ is true if both $A$ and $B$ are true.                                                                                                                  |
| $\forall$                                  | Universal quantifier ("for all").                                                                       | $\forall t \in T$ means for every $t$ in the time window $T$ .                                                                                                      |
| $\exists$                                  | Existential quantifier ("there exists").                                                                | $\exists t \in T$ means for some time $t$ in $T$ .                                                                                                                  |
| <b>Arithmetic and Functional Operators</b> |                                                                                                         |                                                                                                                                                                     |

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Table 1 – continued from previous page

| Symbol                                       | Meaning                                                                  | Example or Notes                                                                                                                                                                               |
|----------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\sum$                                       | Summation operator.                                                      | $\sum_{i=1}^4 A_i = A_1 + A_2 + A_3 + A_4$ .                                                                                                                                                   |
| $\max$                                       | Maximum operator.                                                        | $\max_{t \in T} f(t)$ evaluates $f(t)$ over all $t \in T$ and returns the maximum value. $\max \mathbf{A}$ returns the largest value of the set $\mathbf{A}$ , e.g., $\max \{1, 2, 3\} = 3$ .  |
| $\min$                                       | Minimum operator.                                                        | $\min_{t \in T} f(t)$ evaluates $f(t)$ over all $t \in T$ and returns the minimum value. $\min \mathbf{A}$ returns the smallest value of the set $\mathbf{A}$ , e.g., $\min \{1, 2, 3\} = 1$ . |
| $\times$                                     | Multiplication operator.                                                 |                                                                                                                                                                                                |
| $\mathbb{I}\{A\}$                            | Indicator function returning 1 if condition $A$ is true and 0 otherwise. | Defined in (4).                                                                                                                                                                                |
| <b>Time and Indexing</b>                     |                                                                          |                                                                                                                                                                                                |
| $t$                                          | Time measured in minutes relative to encounter start.                    | Negative values denote pre-encounter measurements, $t = 0$ marks encounter start, and positive values occur during the encounter.                                                              |
| $T$                                          | Time window measured in minutes.                                         | $T = [0, 1440)$ represents the first 24 hours following encounter start.                                                                                                                       |
| $m_{\text{resp}}$                            | Look-back window for respiratory related values                          | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{vaso}}$                            | Look-back window for vasoactive medications                              | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{bp}}$                              | Look-back window for blood pressure measurements                         | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{lac}}$                             | Look-back window for lactate                                             | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{gcs}}$                             | Look-back window for GCS                                                 | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{pupil}}$                           | Look-back window for Pupils                                              | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{coag}}$                            | Look-back window for coagulation related values                          | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{endo}}$                            | Look-back window for endocrine related values                            | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{immu}}$                            | Look-back window for immunologic related values                          | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{hepatic}}$                         | Look-back window for hepatic related values                              | $[0, \infty)$                                                                                                                                                                                  |
| $m_{\text{renal}}$                           | Look-back window for renal related values                                | $[0, \infty)$                                                                                                                                                                                  |
| $\mathbf{m}$                                 | Vector of look-back windows used for carry-forward operations.           |                                                                                                                                                                                                |
| <b>Observation History and Carry-Forward</b> |                                                                          |                                                                                                                                                                                                |
| $\tau_X(t)$                                  | Most recent observation time for variable $X$ at or before time $t$ .    | $\tau_X(t)$ identifies when $X$ was last measured prior to $t$ . Defined in (1).                                                                                                               |
| $\Delta_X(t)$                                | Staleness of measurement for variable $X$ .                              | $\Delta_X(t)$ is the elapsed time since $\tau_X(t)$ . Defined in (2).                                                                                                                          |

Continued on next page

Table 1 – continued from previous page

| Symbol                     | Meaning                                                                                        | Example or Notes                                                                                    |
|----------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $\mathcal{C}[X](t; m)$     | Carry-forward operator returning the most recent value of $X$ within $m$ minutes of time $t$ . | Returns the last observed value of $X$ if $\Delta_X(t) \leq m$ , otherwise NA. Defined in (3).      |
| $\mathcal{D}[x](t; \cdot)$ | Scoring operator: the point contribution of scoring entity $x$ at time $t$ .                   | When the common argument list $(m, \text{Age})$ is used, arguments not relevant to $x$ are ignored. |
| NA                         | Missing, unknown, or undefined value.                                                          | Returned when a required measurement is unavailable.                                                |

Two points are especially important for reading the rest of the document. First,  $:=$  is **not** the same as  $=$ .  $:=$  means we are *defining* something. Second, NA means “missing/unknown/undefined” (e.g., a lab value was not measured), which is different from a numeric value like 0. By contrast,  $\emptyset$  is the **empty set** (no elements), which we use in set-based definitions like  $\max \emptyset = \text{NA}$ .

For arithmetic, when NA is involved, the outcome is always NA, e.g.,  $\text{NA} + X := \text{NA} \forall X$ . Table 2 explicitly defines how logical statements are interpreted when working with NA. When a value is missing, we avoid forcing it into “true” or “false.” The rules below show how missingness propagates through logical statements. If either side of an “and/or” statement is unknown, the whole statement may be unknown unless the other side already determines the answer.

Table 2: Logic with NA.

| Statement                       | Value |
|---------------------------------|-------|
| <b>‘OR’ statements</b>          |       |
| $\text{NA} \vee \text{NA}$      | NA    |
| $\text{NA} \vee \text{true}$    | true  |
| $\text{NA} \vee \text{false}$   | NA    |
| <b>‘And’ statements</b>         |       |
| $\text{NA} \wedge \text{NA}$    | NA    |
| $\text{NA} \wedge \text{true}$  | NA    |
| $\text{NA} \wedge \text{false}$ | false |

## 2.2 Function Definitions

This section defines how we turn time-stamped clinical data into a single value at any time  $t$ . In real EHR data, measurements are taken at irregular times. The definitions below formalize “what is the most recent value?” and “is it too old to trust?” so that scoring is reproducible.

For any time-indexed physiologic variable  $X$ , we write  $X(t)$  for the value recorded at time  $t$  where  $t$  is the time, in minutes from the encounter start, i.e.,  $t = 0$  denotes encounter start,  $t = 32$  is 32 minutes into the encounter, and  $t = -12$  is 12 minutes before encounter start. The negative times represent measurements recorded before encounter start, such as EMS data.

For any variable  $X$ , define the last observation time prior to or at  $t$  as:

$$\tau_X(t) := \max \{s \leq t : X(s) \text{ observed or reported}\}, \quad \max \emptyset = \text{NA}. \quad (1)$$

Define the staleness of the measurement:

$$\Delta_X(t) := \begin{cases} t - \tau_X(t) & \tau_X(t) \neq \text{NA} \\ \text{NA} & \text{otherwise.} \end{cases} \quad (2)$$

We define a staleness-limited carry-forward operator. Carry forward when working with retrospective data, a look back limit when working with prospective or real time data. For  $m \geq 0$  minutes,

$$\mathcal{C}[X](t; m) := \begin{cases} X(\tau_X(t)) & \Delta_X(t) \leq m \\ \text{NA} & \text{otherwise.} \end{cases} \quad (3)$$

We will use an indicator function that returns 1 or 0:

$$\mathbb{I}\{A\} := \begin{cases} 1 & A \text{ is true} \\ 0 & A \text{ is false or NA.} \end{cases} \quad (4)$$

In words:  $\tau_X(t)$  tells you when the last measurement happened,  $\Delta_X(t)$  tells you how old it is, and  $\mathcal{C}[X](t; m)$  returns that last value if it is recent enough (within  $m$  minutes); otherwise it returns “missing.” This is how we prevent using values that are too old to reflect current physiology. The indicator  $\mathbb{I}\{A\}$  then lets us convert logical statements (e.g., “lactate is at least 5”) into 0/1 points for scoring.

For potentially incomplete time-indexed function  $f$ , extrema are evaluated over the available values of  $f$ . Define:

$$\mathcal{A}_f(T) := \{f(t) : t \in T, f(t) \text{ is available}\}. \quad (5)$$

Then

$$\max_{t \in T} f(t) := \begin{cases} \max \mathcal{A}_f(T) & \mathcal{A}_f(T) \neq \emptyset \\ \text{NA} & \mathcal{A}_f(T) = \emptyset, \end{cases} \quad (6)$$

and

$$\min_{t \in T} f(t) := \begin{cases} \min \mathcal{A}_f(T) & \mathcal{A}_f(T) \neq \emptyset \\ \text{NA} & \mathcal{A}_f(T) = \emptyset. \end{cases} \quad (7)$$

### 3 Defining Organ System Dysfunction Scores

We assess eight organ systems: respiratory, cardiovascular, neurologic, coagulation, endocrine, immunologic, hepatic, and renal. Each subsection below lists the EHR inputs needed to compute that organ score, then shows how those inputs are transformed into constructed variables and point assignments. The goal is to make every step of the scoring reproducible: given the same time-stamped data, you should arrive at the same score.

The listed patient, laboratory, observation, event, and medication fields are as we expect them to appear in the electronic health record. These are inputs, not judgments about clinical severity. We define a domain for the variable. These domains are intentionally broad (wider than clinically plausible values) to keep data validation separate from scoring.

#### 3.1 Respiratory

##### 3.1.1 Data from the EHR

There are several values we expect to get from the Electronic Health Record capturing oxygenation,  $\text{FiO}_2$  (fraction of inspired oxygen),  $\text{SpO}_2$  (pulse oximetry oxygen saturation), and  $\text{PaO}_2$  (arterial oxygen pressure). Additionally, the presence of mechanical or noninvasive respiratory support. Together, they allow us to quantify how well the patient is oxygenating and whether that oxygenation depends on advanced support.

Table 3: EHR Inputs for Respiratory Organ Dysfunction Scoring

| Variable       | Description                 | Type | Units    | Domain       |
|----------------|-----------------------------|------|----------|--------------|
| $\text{FiO}_2$ | Fraction of Inspired Oxygen | Real | unitless | [0.21, 1.00] |

Table 3 – continued from previous page

| Variable                        | Description                                   | Type    | Units               | Domain                 |
|---------------------------------|-----------------------------------------------|---------|---------------------|------------------------|
| SpO <sub>2</sub>                | Pulse oximetry oxygen saturation              | Integer | unitless            | $\{0, 1, \dots, 100\}$ |
| PaO <sub>2</sub>                | Arterial oxygen pressure                      | Real    | mmHg                | $[0, \infty)$          |
| P <sub>aw</sub> <sup>VENT</sup> | Mean Airway Pressure, from a ventilator       | Real    | cm H <sub>2</sub> O | $[0, \infty)$          |
| P <sub>aw</sub> <sup>HFOV</sup> | Mean Airway Pressure, from a HFOV             | Real    | cm H <sub>2</sub> O | $[0, \infty)$          |
| P <sub>aw</sub> <sup>PEEP</sup> | Positive end-expiratory pressure              | Real    | cm H <sub>2</sub> O | $[0, \infty)$          |
| VENT                            | Indicator for invasive mechanical ventilation | Integer |                     | $\{0, 1\}$             |
| O <sub>2</sub> Support          | Any non-invasive oxygen support               | Integer |                     | $\{0, 1\}$             |

### 3.1.2 Constructed Variables

From the EHR values we construct several variables which are then used for the respiratory score. The constructed variables combine measurements that are not always recorded at the same time (e.g., SpO<sub>2</sub> and FiO<sub>2</sub>) and apply clinical logic about timing and support. This keeps us from pairing an outdated FiO<sub>2</sub> with a more recent oxygen measurement or vice versa.

**PF Ratio** For determining the PF ratio at time  $t$  we consider look-back windows for both FiO<sub>2</sub> and PaO<sub>2</sub> and the relative look-back between the two. Specifically, the PFR is only valid when

$$\tau_{\text{FiO}_2}(t) \leq \tau_{\text{PaO}_2}(t) \quad \equiv \quad \Delta_{\text{FiO}_2}(t) \geq \Delta_{\text{PaO}_2}(t). \quad (8)$$

Conceptually, this restriction is in place to protect from using a PaO<sub>2</sub> value which was taken *before* the most current FiO<sub>2</sub> value. That is, PFR is only valid if the PaO<sub>2</sub> value is reported at the same time, or after the FiO<sub>2</sub> is reported.

The PFR at time  $t$  is:

$$\text{PFR}(t; m_{\text{resp}}) := \begin{cases} \frac{\mathcal{C}[\text{PaO}_2](t; m_{\text{resp}})}{\mathcal{C}[\text{FiO}_2](t; m_{\text{resp}})} & \tau_{\text{FiO}_2}(t) \leq \tau_{\text{PaO}_2}(t) \\ \text{NA} & \text{otherwise.} \end{cases} \quad (9)$$

With this definition, at time  $t$ , the PF ratio will either have a valid numeric value or be NA.

**SF Ratio** Like the PF ratio, we only consider the SF ratio valid if the SpO<sub>2</sub> value is reported at the same time as the FiO<sub>2</sub> or after.

$$\tau_{\text{FiO}_2}(t) \leq \tau_{\text{SpO}_2}(t) \quad \equiv \quad \Delta_{\text{FiO}_2}(t) \geq \Delta_{\text{SpO}_2}(t). \quad (10)$$

Additionally, the SF ratio is valid if the SpO<sub>2</sub>  $\leq 97$ . This requirement is based on the PODIUM [Bembea et al., 2022, Squires et al., 2022] organ dysfunction scoring, and is an additional requirement on the pSOFA [Matics and Sanchez-Pinto, 2017] cut points in the scoring.

$$\text{SFR}(t; m_{\text{resp}}) := \begin{cases} \frac{\mathcal{C}[\text{SpO}_2](t; m_{\text{resp}})}{\mathcal{C}[\text{FiO}_2](t; m_{\text{resp}})} & \tau_{\text{FiO}_2}(t) \leq \tau_{\text{SpO}_2}(t) \wedge \mathcal{C}[\text{SpO}_2](t; m_{\text{resp}}) \leq 97 \\ \text{NA} & \text{otherwise.} \end{cases} \quad (11)$$

**Invasive Mechanical Ventilation** We operationalize the definition for invasive mechanical ventilation based in part of differences in the EHR records we had access to. In some sets, a simple VENT indicator variable was provided. In other EHR sets we had mean airway pressure from either a ventilator or high-frequency oscillatory ventilation (HFOV). Additionally, we had positive end-expiratory pressure (PEEP) to consider as well.

We defined invasive mechanical ventilation (IMV) as an indicator based on four possible conditions.

$$\text{IMV}(t; m_{\text{resp}}) := \mathbb{I} \left\{ \bigvee_{i=1}^4 A_i(t; m_{\text{resp}}) \right\} \quad (12)$$

where

$$\begin{aligned} A_1(t; m_{\text{resp}}) &:= \mathcal{C}[\text{VENT}](t; m_{\text{resp}}) = 1, \\ A_2(t; m_{\text{resp}}) &:= \mathcal{C}[\text{P}_{\text{aw}}^{\text{VENT}}](t; m_{\text{resp}}) > 0, \\ A_3(t; m_{\text{resp}}) &:= \mathcal{C}[\text{P}_{\text{aw}}^{\text{HFOV}}](t; m_{\text{resp}}) > 0, \text{ and} \\ A_4(t; m_{\text{resp}}) &:= \mathcal{C}[\text{P}_{\text{aw}}^{\text{PEEP}}](t; m_{\text{resp}}) > 3. \end{aligned} \quad (13)$$

The PEEP limit of 3 cm H<sub>2</sub>O is used to limit false positives while limiting false negatives. Similar limits have been used in other criteria. For example, the lung injury score [Martin and Bernard, 2001] uses a PEEP of 5 cm H<sub>2</sub>O for 0 points and 6-8 cm H<sub>2</sub>O. Further, Martin and Bernard [2001] states a PEEP of 5 cm H<sub>2</sub>O and FiO<sub>2</sub> < 0.5 as part of the criteria for identifying patients ready to accept a trial of spontaneous breathing.

**Other Respiratory Support** Extend the definitions in (13) to include

$$\begin{aligned} A_5(t; m_{\text{resp}}) &:= \mathcal{C}[\text{FiO}_2](t; m_{\text{resp}}) > 0.21, \\ A_6(t; m_{\text{resp}}) &:= \mathcal{C}[\text{O}_2\text{Support}](t; m_{\text{resp}}) = 1. \end{aligned} \quad (14)$$

Then other respiratory support is defined by (13) and (14) as

$$\text{ORS}(t; m_{\text{resp}}) := \mathbb{I} \left\{ \bigvee_{i=1}^6 A_i(t; m_{\text{resp}}) \right\} \quad (15)$$

### 3.1.3 Respiratory Organ Dysfunction Score

In the development of Phoenix, we used pSOFA respiratory scoring [Matics and Sanchez-Pinto, 2017]. The pSOFA respiratory scoring does not explicitly require respiratory support to score 1 point, however, for the Phoenix definition the requirement of some respiratory support was made explicit. It is extremely unlikely that a patient with SFR or PFR values sufficiently low enough to score at least one point for respiratory dysfunction would not have at the very least supplemental oxygen.

$$\mathcal{D}[\text{Resp}](t; m_{\text{resp}}) := \text{ORS}(t; m_{\text{resp}}) \mathbb{I}\{B_1(t; m_{\text{resp}})\} + \text{IMV}(t; m_{\text{resp}}) \sum_{i=2}^3 \mathbb{I}\{B_i(t; m_{\text{resp}})\} \quad (16)$$

where

$$\begin{aligned} B_1(t; m_{\text{resp}}) &:= \text{PFR}(t; m_{\text{resp}}) < 400 \vee \text{SFR}(t; m_{\text{resp}}) < 292 \\ B_2(t; m_{\text{resp}}) &:= \text{PFR}(t; m_{\text{resp}}) < 200 \vee \text{SFR}(t; m_{\text{resp}}) < 220 \\ B_3(t; m_{\text{resp}}) &:= \text{PFR}(t; m_{\text{resp}}) < 100 \vee \text{SFR}(t; m_{\text{resp}}) < 148. \end{aligned} \quad (17)$$

The domain for the respiratory organ dysfunction score is

$$\mathcal{D}[\text{Resp}](t; m_{\text{resp}}) \in \{0, 1, 2, 3\} \quad \forall m_{\text{resp}}, t. \quad (18)$$

## 3.2 Cardiovascular

The cardiovascular score has three components, vasoactive medications, mean arterial pressure, and lactate. A score of 0, 1, or 2 points is possible for each of the components resulting in a total score of 0 to 6 points.



### 3.2.1 Data from the EHR

The input data from the EHR needed for the cardiovascular dysfunction scores are vasoactive medications (as indicators of pharmacologic cardiovascular support), blood pressure measurements (from cuff or arterial line), lactate, and age (for age-specific mean arterial pressure thresholds).

Table 4: EHR Inputs for Cardiovascular Organ Dysfunction Scoring

| Variable         | Description                                          | Type    | Units    | Domain        |
|------------------|------------------------------------------------------|---------|----------|---------------|
| Dobutamine       | Indicator of systemic use of vasoactive medication   | Integer | –        | $\{0, 1\}$    |
| Dopamine         | Indicator of systemic use of vasoactive medication   | Integer | –        | $\{0, 1\}$    |
| Epinephrine      | Indicator of systemic use of vasoactive medication   | Integer | –        | $\{0, 1\}$    |
| Milrinone        | Indicator of systemic use of vasoactive medication   | Integer | –        | $\{0, 1\}$    |
| Norepinephrine   | Indicator of systemic use of vasoactive medication   | Integer | –        | $\{0, 1\}$    |
| Vasopressin      | Indicator of systemic use of vasoactive medication   | Integer | –        | $\{0, 1\}$    |
| Lactate          | Lactate                                              | Real    | mmol / L | $[0, \infty)$ |
| SBP <sub>c</sub> | systolic blood pressure reported from a cuff         | Real    | mmHg     | $[0, \infty)$ |
| DBP <sub>c</sub> | diastolic blood pressure reported from a cuff        | Real    | mmHg     | $[1, 200]$    |
| MAP <sub>c</sub> | mean arterial pressure reported from a cuff          | Real    | mmHg     | $[0, \infty)$ |
| SBP <sub>a</sub> | systolic blood pressure reported from arterial line  | Real    | mmHg     | $[0, \infty)$ |
| DBP <sub>a</sub> | diastolic blood pressure reported from arterial line | Real    | mmHg     | $[1, 200]$    |
| MAP <sub>a</sub> | mean arterial pressure reported from arterial line   | Real    | mmHg     | $[0, \infty)$ |
| Age              | Patient’s age                                        | Real    | months   | $[0, 216]$    |

### 3.2.2 Constructed Variable

We build three components: vasoactive medication use, mean arterial pressure, and lactate. Each component contributes 0–2 points, and together they sum to the cardiovascular score. The definitions below formalize how we combine multiple data sources (e.g., cuff vs arterial line) and how we handle timing.

**Vasoactive Medications** Each of the six vasoactive medications

$$V := \{\text{Dobutamine, Dopamine, Epinephrine, Milrinone, Norepinephrine, Vasopressin}\} \quad (19)$$

are to be assessed as 0/1 indicators. Either the patient is receiving systemic vasoactive medication for the purpose of cardiovascular support or they are not. Medications such as epinephrine are used in many different situations for different purposes, it is important that the input data is an indicator of the use of the medication systemically for cardiovascular support.

We define the contribution to the cardiovascular score from vasoactive medications as the count of the number of vasoactive medications received, regardless of dose, with a cap of 2 points.

$$\mathcal{D}[\text{Vasos}](t; m_{\text{vaso}}) := \min \left\{ 2, \sum_{v \in V} \mathbb{I} \{ \mathcal{C}[v](t; m_{\text{vaso}}) = 1 \} \right\}. \quad (20)$$

The domain for the contribution to the cardiovascular score from vasoactive medications is:

$$\mathcal{D}[\text{Vasos}](t; m_{\text{vaso}}) \in \{0, 1, 2\} \quad \forall m_{\text{vaso}}, t \quad (21)$$

This was based on the use of the vasoactive inotrope score (VIS)[[Haque et al., 2015](#)]. The VIS used the same six vasoactive medications and based the score on the dosage. During the development of the Phoenix criteria we found some lower resourced sites did not have all six medications. As such, patients at these

lower resourced sites had a ceiling on the severity of cardiovascular dysfunction based on VIS. By ignoring the dosage and just relying on the fact that the patient was receiving the medication, the Phoenix criteria was determined to have greater utility across high-, medium-, and low- resourced sites.

**Mean Arterial Pressure (MAP)** Mean arterial pressure (MAP) is non-trivial to define. We have limits based on age (in months) and a hierarchy for the preferable source of information used to define the MAP. In practice, MAP may be recorded directly or inferred from systolic and diastolic pressures. We prioritize arterial line measurements when available because they are more reliable in unstable patients, then fall back to cuff measurements.

The hierarchy for the pressure: use the MAP from an arterial line ( $\text{MAP}_a$ ) first. If that is not available, then approximate using systolic and diastolic values reported from the arterial line.  $\text{SBP}_a$  and  $\text{DBP}_a$ . If none of the arterial line values are available, then use values from a cuff preferentially taking a reported mean arterial pressure,  $\text{MAP}_c$ , and lastly, approximate via  $\text{SBP}_c$  and  $\text{DBP}_c$ . That is, let

$$\begin{aligned}\widehat{\text{MAP}}_1(t; m_{bp}) &:= \mathcal{C}[\text{MAP}_a](t; m_{bp}), \\ \widehat{\text{MAP}}_2(t; m_{bp}) &:= \frac{1}{3}\mathcal{C}[\text{SBP}_a](t; m_{bp}) + \frac{2}{3}\mathcal{C}[\text{DBP}_a](t; m_{bp}), \\ \widehat{\text{MAP}}_3(t; m_{bp}) &:= \mathcal{C}[\text{MAP}_c](t; m_{bp}), \text{ and} \\ \widehat{\text{MAP}}_4(t; m_{bp}) &:= \frac{1}{3}\mathcal{C}[\text{SBP}_c](t; m_{bp}) + \frac{2}{3}\mathcal{C}[\text{DBP}_c](t; m_{bp}).\end{aligned}\tag{22}$$

Then

$$\text{MAP}^*(t; m_{bp}) := \begin{cases} \widehat{\text{MAP}}_{i^*}(t; m_{bp}) & \text{if } \exists i^* \in \{1, 2, 3, 4\} \text{ where} \\ \text{NA} & i^* := \min \{i \in \{1, 2, 3, 4\} : \widehat{\text{MAP}}_i(t; m_{bp}) \neq \text{NA}\} \\ & \text{otherwise.} \end{cases}\tag{23}$$

The limits for the MAP are based on age (in months). Age-based thresholds reflect pediatric physiology: younger patients have lower expected MAPs, so the cut-points increase with age. Let

$$\theta_1(\text{Age}) := \begin{cases} 31 & \text{Age} \in [0, 1) \\ 39 & \text{Age} \in [1, 12) \\ 44 & \text{Age} \in [12, 24) \\ 45 & \text{Age} \in [24, 60) \\ 49 & \text{Age} \in [60, 144) \\ 52 & \text{Age} \in [144, 216), \end{cases}\tag{24}$$

and

$$\theta_2(\text{Age}) := \begin{cases} 17 & \text{Age} \in [0, 1) \\ 25 & \text{Age} \in [1, 12) \\ 31 & \text{Age} \in [12, 24) \\ 32 & \text{Age} \in [24, 60) \\ 36 & \text{Age} \in [60, 144) \\ 38 & \text{Age} \in [144, 216). \end{cases}\tag{25}$$

Then, the part of the cardiovascular score based on MAP is:

$$\mathcal{D}[\text{MAP}](t; m_{bp}, \text{Age}) := \mathbb{I}\{\text{MAP}^*(t; m_{bp}) < \theta_2(\text{Age})\} + \mathbb{I}\{\text{MAP}^*(t; m_{bp}) < \theta_1(\text{Age})\}.\tag{26}$$

The domain for this component of the cardiovascular dysfunction score is

$$\mathcal{D}[\text{MAP}](t; m_{bp}, \text{Age}) \in \{0, 1, 2\} \quad \forall m_{bp}, t.\tag{27}$$

**Lactate** Lactate is treated as a marker of impaired perfusion and severity of shock. Higher lactate thresholds contribute more points. The definition for lactate used in the cardiovascular dysfunction score is:

$$\mathcal{D}[\text{Lactate}](t; m_{\text{lac}}) := \mathbb{I}\{\mathcal{C}[\text{Lactate}](t; m_{\text{lac}}) \geq 5\} + \mathbb{I}\{\mathcal{C}[\text{Lactate}](t; m_{\text{lac}}) \geq 11\}. \quad (28)$$

That is, one point for a lactate value of at least 5 mmol/L; two points for a lactate value of at least 11 mmol/L. The domain for the contribution of lactate to the cardiovascular dysfunction score is:

$$\mathcal{D}[\text{Lactate}](t; m_{\text{lac}}) \in \{0, 1, 2\} \quad \forall m_{\text{lac}}, t. \quad (29)$$

### 3.2.3 Cardiovascular Dysfunction Score

The cardiovascular dysfunction is based on (20), (26), and (28).

$$\mathcal{D}[\text{Card}](t; m_{\text{vaso}}, m_{\text{bp}}, m_{\text{lac}}, \text{Age}) := \mathcal{D}[\text{Vasos}](t; m_{\text{vaso}}) + \mathcal{D}[\text{MAP}](t; m_{\text{bp}}, \text{Age}) + \mathcal{D}[\text{Lactate}](t; m_{\text{lac}}). \quad (30)$$

The domain for the cardiovascular dysfunction score is

$$\mathcal{D}[\text{Card}](t; m_{\text{vaso}}, m_{\text{bp}}, m_{\text{lac}}, \text{Age}) \in \{0, 1, 2, 3, 4, 5, 6\} \quad \forall m, t; \quad (31)$$

## 3.3 Neurological

The neurological score is based on the Glasgow Coma score (GCS) and whether or not the patient has bilaterally fixed pupils. These capture level of consciousness and brainstem function, respectively.

### 3.3.1 Data from the EHR

Table 5: EHR Inputs for Neurological Organ Dysfunction Scoring

| Variable              | Description               | Type    | Units    | Domain             |
|-----------------------|---------------------------|---------|----------|--------------------|
| GCS <sub>motor</sub>  | Glasgow Coma Motor Score  | Integer | unitless | {1, 2, 3, 4, 5, 6} |
| GCS <sub>eye</sub>    | Glasgow Coma Eye Score    | Integer | unitless | {1, 2, 3, 4}       |
| GCS <sub>verbal</sub> | Glasgow Coma Verbal Score | Integer | unitless | {1, 2, 3, 4, 5}    |
| GCS <sub>total</sub>  | Total Glasgow Coma Score  | Integer | unitless | {3, 4, ..., 15}    |
| Pupils                | Bilaterally fixed pupils  | Integer |          | {0, 1}             |

### 3.3.2 Constructed Variable

We derive two variables: a time-aligned GCS total (which may need to be reconstructed from components) and a binary indicator for fixed pupils. This helps address common EHR issues where component scores are updated without the total.

**GCS** In general the GCS total is the sum of the three components, eye, verbal, and motor. In the development of the Phoenix criteria we found that there were cases where one or two of the components would be updated but the other component(s) and the total were not updated. When this occurred we chose to assume it was because there was no change in the other component scores. This rule prevents missing totals from artificially lowering the score when only one component is updated.

$$GCS(t; m_{gcs}) := \begin{cases} \mathcal{C}[GCS_{total}](t; m_{gcs}) & \Delta_{GCS_{total}}(t) < \min_{i \in \mathbf{G}} \Delta_{GCS_i}(t) \\ \sum_{i \in \mathbf{G}} \mathcal{C}[GCS_i](t; \infty) & \left[ \Delta_{GCS_{total}}(t) = \text{NA} \vee \min_{i \in \mathbf{G}} \Delta_{GCS_i}(t) \leq \Delta_{GCS_{total}}(t) \right] \wedge \\ & \min_{i \in \mathbf{G}} \Delta_{GCS_i}(t) \leq m_{gcs} \\ \text{NA} & \text{otherwise,} \end{cases} \quad (32)$$

where

$$\mathbf{G} := \{\text{motor, eye, verbal}\}. \quad (33)$$

The use of  $\mathcal{C}[\cdot](t; \infty)$  means “carry forward the most recent value with no time limit.” In practice, this is only applied when each of the three component scores has been observed at least once. So if the latest eye, verbal, and motor scores were recorded at different times, we take each component’s most recent value and sum them to form a consistent total, even if those component timestamps are not identical.

The score contribution from GCS is

$$\mathcal{D}[GCS](t; m_{gcs}) := \mathbb{I}\{GCS(t; m_{gcs}) \leq 10\} \quad (34)$$

**Pupils** This is treated as a severe neurologic finding and can dominate the neurologic score. The bilaterally fixed pupils are defined as:

$$\mathcal{D}[Pupils](t; m_{pupil}) := 2 \times \mathbb{I}\{\mathcal{C}[Pupils](t; m_{pupil}) = 1\}. \quad (35)$$

### 3.3.3 Neurological Organ Dysfunction Score

The neurological component on the score is:

$$\mathcal{D}[Neuro](t; m_{gcs}, m_{pupil}) := \max\{\mathcal{D}[GCS](t; m_{gcs}), \mathcal{D}[Pupils](t; m_{pupil})\}. \quad (36)$$

## 3.4 Coagulation

The coagulation score uses common laboratory markers of clotting and fibrinolysis. Each marker contributes at most one point, and the total is capped at 2. This keeps the score sensitive to abnormalities without overweighting multiple abnormal labs that can occur together in the same patient.

For readers without a clinical background: Clotting (coagulation) is the body’s process for stopping bleeding by forming fibrin clots. Fibrinolysis is the counter-process that breaks down those clots once they are no longer needed. INR is a standardized measure of how long it takes blood to clot; higher INR means slower clotting and can reflect coagulopathy, liver dysfunction, or anticoagulation. D-dimer is a breakdown product of cross-linked fibrin, so elevated levels suggest recent or ongoing clot formation and breakdown, though it is not specific to any single disease.

In severe infection, clotting can be activated even without trauma. Inflammatory signals and endothelial injury can trigger coagulation pathways, leading to micro-clot formation in small vessels. At the same time, fibrinolysis can be overwhelmed or dysregulated. In extreme cases this becomes disseminated intravascular coagulation (DIC), where clotting and bleeding risk coexist.

### 3.4.1 Data from the EHR

Table 6: EHR Inputs for Coagulation Organ Dysfunction Scoring

| Variable   | Description | Type | Units                | Domain        |
|------------|-------------|------|----------------------|---------------|
| Platelets  | Platelets   | Real | 1000 / $\mu\text{L}$ | $[0, \infty)$ |
| INR        | INR         | Real | unitless             | $[0, \infty)$ |
| DDimer     | D-Dimer     | Real | mg/L (FEU)           | $[0, \infty)$ |
| Fibrinogen | Fibrinogen  | Real | mg/dL                | $[0, \infty)$ |

### 3.4.2 Constructed Variable

There are no variables to construct.

### 3.4.3 Coagulation Organ Dysfunction Score

Each indicator below tests whether a lab value crosses a clinically relevant threshold. The score sums how many thresholds are met, then caps at 2 points.

Define:

$$\begin{aligned}
\mathcal{D}[\text{Platelets}](t; m_{\text{coag}}) &:= \mathbb{I}\{\mathcal{C}[\text{Platelets}](t; m_{\text{coag}}) < 100\} \\
\mathcal{D}[\text{INR}](t; m_{\text{coag}}) &:= \mathbb{I}\{\mathcal{C}[\text{INR}](t; m_{\text{coag}}) > 1.3\} \\
\mathcal{D}[\text{DDimer}](t; m_{\text{coag}}) &:= \mathbb{I}\{\mathcal{C}[\text{DDimer}](t; m_{\text{coag}}) > 2\} \\
\mathcal{D}[\text{Fibrinogen}](t; m_{\text{coag}}) &:= \mathbb{I}\{\mathcal{C}[\text{Fibrinogen}](t; m_{\text{coag}}) < 100\},
\end{aligned} \tag{37}$$

and then

$$\mathcal{D}[\text{Coag}](t; m_{\text{coag}}) := \min \left\{ 2, \sum_{x \in \mathcal{C}_{\text{coag}}} \mathcal{D}[x](t; m_{\text{coag}}) \right\}, \tag{38}$$

where

$$\mathcal{C}_{\text{coag}} := \{\text{Platelets}, \text{INR}, \text{DDimer}, \text{Fibrinogen}\}. \tag{39}$$

## 3.5 Endocrine

The endocrine score is part of the Phoenix-8 scoring system.

### 3.5.1 Data from the EHR

There is only one element from the EHR used in the endocrine dysfunction assessment: glucose, Table 7.

Table 7: EHR Inputs for Endocrine Organ Dysfunction Scoring

| Variable | Description   | Type | Units | Domain        |
|----------|---------------|------|-------|---------------|
| Glucose  | Blood Glucose | Real | mg/dL | $[0, \infty)$ |

### 3.5.2 Constructed Variables

There are no constructed variables to define.

### 3.5.3 Endocrine Organ Dysfunction Score

$$\mathcal{D}[\text{Endo}](t; m_{\text{endo}}) := \mathbb{I}\{\mathcal{C}[\text{Glucose}](t; m_{\text{endo}}) < 50 \vee \mathcal{C}[\text{Glucose}](t; m_{\text{endo}}) > 150\} \tag{40}$$

The domain for the endocrine score is

$$\mathcal{D}[\text{Endo}](t; m_{\text{endo}}) \in \{0, 1\}, \quad \forall m_{\text{endo}}, t. \tag{41}$$

### 3.6 Immunologic

The immunologic score is part of the Phoenix-8 scoring system.

#### 3.6.1 Data from the EHR

Two values are required from the EHR: Absolute neutrophil count (ANC) and absolute lymphocyte count (ALC). Both are expected in units of 1000 cells per cubic millimeter, Table 8.

Table 8: EHR Inputs for Immunologic Organ Dysfunction Scoring

| Variable | Description               | Type | Units                        | Domain |
|----------|---------------------------|------|------------------------------|--------|
| ANC      | Absolute neutrophil count | Real | 1000 cells / mm <sup>3</sup> | [0, ∞) |
| ALC      | Absolute lymphocyte count | Real | 1000 cells / mm <sup>3</sup> | [0, ∞) |

#### 3.6.2 Constructed Variables

There are no constructed variables for the immunologic score.

#### 3.6.3 Immunologic Organ Dysfunction Score

Define

$$\mathcal{D}[\text{ALC}](t; m_{\text{immu}}) := \mathbb{I}\{\mathcal{C}[\text{ALC}](t; m_{\text{immu}}) < 1.00\}, \quad (42)$$

and

$$\mathcal{D}[\text{ANC}](t; m_{\text{immu}}) := \mathbb{I}\{\mathcal{C}[\text{ANC}](t; m_{\text{immu}}) < 0.50\}. \quad (43)$$

Then

$$\mathcal{D}[\text{Immu}](t; m_{\text{immu}}) := \max\{\mathcal{D}[\text{ALC}](t; m_{\text{immu}}), \mathcal{D}[\text{ANC}](t; m_{\text{immu}})\}. \quad (44)$$

The domain for the immunologic score is

$$\mathcal{D}[\text{Immu}](t; m_{\text{immu}}) \in \{0, 1\}, \quad \forall m_{\text{immu}}, t. \quad (45)$$

### 3.7 Hepatic

#### 3.7.1 Data from the EHR

Two values are required from the EHR: total bilirubin reported in units of mg/dL and alanine aminotransferase (ALT) reported in units of IU/L, Table 9.

Table 9: EHR Inputs for Hepatic Organ Dysfunction Scoring

| Variable  | Description              | Type | Units  | Domain |
|-----------|--------------------------|------|--------|--------|
| Bilirubin | Total bilirubin          | Real | mg/dL  | [0, ∞) |
| ALT       | Alanine aminotransferase | Real | IU / L | [0, ∞) |

#### 3.7.2 Constructed Variables

There are no constructed variables for the hepatic organ dysfunction assessment.

### 3.7.3 Hepatic Organ Dysfunction Score

As with endocrine and immunologic scores, the hepatic score can be a 0 or 1 from either the bilirubin or ALT values.

Define:

$$\mathcal{D}[\text{Bilirubin}](t; m_{\text{hepatic}}) := \mathbb{I}\{\mathcal{C}[\text{Bilirubin}](t; m_{\text{hepatic}}) \geq 4\}, \quad (46)$$

and

$$\mathcal{D}[\text{ALT}](t; m_{\text{hepatic}}) := \mathbb{I}\{\mathcal{C}[\text{ALT}](t; m_{\text{hepatic}}) > 102\}. \quad (47)$$

Then

$$\mathcal{D}[\text{Hepatic}](t; m_{\text{hepatic}}) := \max\{\mathcal{D}[\text{Bilirubin}](t; m_{\text{hepatic}}), \mathcal{D}[\text{ALT}](t; m_{\text{hepatic}})\}. \quad (48)$$

The domain for the hepatic score is

$$\mathcal{D}[\text{Hepatic}](t; m_{\text{hepatic}}) \in \{0, 1\}, \quad \forall m_{\text{hepatic}}, t. \quad (49)$$

## 3.8 Renal

The renal score is part of the Phoenix-8 scoring system.

### 3.8.1 Data from the EHR

The patient's age (at admission, in months) and their creatinine levels (in mg/dL) are required to assess renal dysfunction. Age is also required for assessment of cardiovascular dysfunction as part of the Phoenix scoring and thus the only additional variable to collect to extend to Phoenix-8 from Phoenix for the renal dysfunction score is the creatinine.

Table 10: EHR Inputs for Renal Organ Dysfunction Scoring

| Variable   | Description   | Type | Units   | Domain        |
|------------|---------------|------|---------|---------------|
| Creatinine | Creatinine    | Real | mg / dL | $[0, \infty)$ |
| Age        | Patient's age | Real | months  | $[0, 216)$    |

### 3.8.2 Constructed Variables

There are no constructed variables required for the assessment of renal dysfunction.

### 3.8.3 Renal Organ Dysfunction Score

The renal dysfunction score is based on an age adjusted creatinine level (50). The cut points for the age ranges are the same as used in the cardiovascular scoring, (24) and (25).

$$\mathcal{D}[\text{Renal}](t; m_{\text{renal}}, \text{Age}) := \begin{cases} 1 & \text{Age} \in [0, 1) \wedge \mathcal{C}[\text{Creatinine}](t; m_{\text{renal}}) \geq 0.8 \\ 1 & \text{Age} \in [1, 12) \wedge \mathcal{C}[\text{Creatinine}](t; m_{\text{renal}}) \geq 0.3 \\ 1 & \text{Age} \in [12, 24) \wedge \mathcal{C}[\text{Creatinine}](t; m_{\text{renal}}) \geq 0.4 \\ 1 & \text{Age} \in [24, 60) \wedge \mathcal{C}[\text{Creatinine}](t; m_{\text{renal}}) \geq 0.6 \\ 1 & \text{Age} \in [60, 144) \wedge \mathcal{C}[\text{Creatinine}](t; m_{\text{renal}}) \geq 0.7 \\ 1 & \text{Age} \in [144, 216) \wedge \mathcal{C}[\text{Creatinine}](t; m_{\text{renal}}) \geq 1.0 \\ 0 & \text{otherwise.} \end{cases} \quad (50)$$

### 3.9 Suspected Infection

A patient is defined to have a suspected infection if they have at least one infectious test *ordered* and at least one dose of a systemic anti-infective agent. Results of the tests are not needed, just that the tests have been ordered. Anti-infective agents include anti-bacterials, anti-fungals, anti-virals, or anti-mycobacterial (anti-buberculosis). This definition is operational: it identifies when clinicians treated the case as infectious, not whether infection was ultimately confirmed.

Let  $t_1$  denote the time when an infectious test was ordered. Let  $t_2$  denote the time when the first dose of a systemic anti-infectious agent was administered.

Suspected infection status is defined as

$$\text{SI}(t) := \mathbb{I}\{t_1 \leq t\} \times \mathbb{I}\{t_2 \leq t\}. \quad (51)$$

## 4 Phoenix Sepsis Score

Define the look-back windows as the vector

$$\mathbf{m} := (m_{\text{resp}}, m_{\text{vaso}}, m_{\text{bp}}, m_{\text{lac}}, m_{\text{gcs}}, m_{\text{pupil}}, m_{\text{coag}}, m_{\text{endo}}, m_{\text{immu}}, m_{\text{hepatic}}, m_{\text{renal}}). \quad (52)$$

For notational compactness,  $\mathcal{D}[x](t; \mathbf{m}, \text{Age})$  uses a common argument list for all scoring entities  $x$ . Each scoring function depends only on the elements of  $\mathbf{m}$ , and on Age, that are relevant to its definition; all other arguments are ignored.

In this definition, the time interval of interest is the same for both SI and Phoenix Sepsis Score (PSS), but the maxima are taken separately. That is, a patient can have  $\text{SI}(t) = 1$  at one time and the maximum organ dysfunction score at another. The Phoenix Sepsis Score combines organ dysfunction with evidence of suspected infection (via SI), so it reflects organ dysfunction *in the presence of* suspected infection.

$$\text{PSS}_d(T; \mathbf{m}, \text{Age}) := \left[ \max_{t \in T} \text{SI}(t) \right] \left[ \max_{t \in T} \sum_{x \in \Omega_d} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \right], \quad d \in \{4, 8\}, \quad (53)$$

where

$$\Omega_4 := \{\text{Resp}, \text{Card}, \text{Neuro}, \text{Coag}\} \quad (54)$$

and

$$\Omega_8 := \Omega_4 \cup \{\text{Endo}, \text{Immu}, \text{Hepatic}, \text{Renal}\}. \quad (55)$$

Sepsis is defined as organ-dysfunction in the presence of a suspected infection. Specifically, a score of at least  $\sigma > 0$  points:

$$\text{Sepsis}(T; \mathbf{m}, \text{Age}) := \mathbb{I}\{\text{PSS}_4(T; \mathbf{m}, \text{Age}) \geq \sigma\}. \quad (56)$$

Septic Shock is sepsis with at least  $\kappa$  point(s) from cardiovascular dysfunction.

$$\text{SepticShock}(T; \mathbf{m}, \text{Age}) := \mathbb{I}\left\{ \max_{t \in T} \text{SI}(t) \times \max_{t \in T} \left( \sum_{x \in \Omega_4} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \mathbb{I}\{\mathcal{D}[\text{Card}](t; \mathbf{m}, \text{Age}) \geq \kappa\} \right) \geq \sigma \right\}, \quad (57)$$

where  $\sigma \geq \kappa > 0$ . Note that the specific shock definition does not simply consider any cardiovascular dysfunction in the window  $T$ . It must occur with the organ dysfunction associated with reaching the sepsis score threshold.

For the published Phoenix Sepsis Criteria [Sanchez-Pinto et al., 2024, Schlappbach et al., 2024] the following parameter values in Table 11 were used:



Table 11: Parameter values defining time windows, look-backs, and thresholds for the Phoenix Sepsis Score, Sepsis, and Septic Shock.

| Parameter            | Value        | Notes                               |
|----------------------|--------------|-------------------------------------|
| <b>Time Window</b>   |              |                                     |
| $T$                  | $[0, 1440)$  | The first 24-hours of the encounter |
| <b>Thresholds</b>    |              |                                     |
| $\sigma$             | 2            |                                     |
| $\kappa$             | 1            |                                     |
| <b>Look-backs</b>    |              |                                     |
| $m_{\text{resp}}$    | 360 minutes  | all respiratory EHR variables       |
| $m_{\text{vaso}}$    | 720 minutes  | vasoactive medications              |
| $m_{\text{bp}}$      | 360 minutes  | blood pressure / MAP                |
| $m_{\text{lac}}$     | 360 minutes  | lactate                             |
| $m_{\text{gcs}}$     | 720 minutes  | GCS components or total             |
| $m_{\text{pupil}}$   | 720 minutes  | pupils                              |
| $m_{\text{coag}}$    | 1440 minutes | coagulation labs                    |
| $m_{\text{endo}}$    | 720 minutes  | glucose                             |
| $m_{\text{immu}}$    | 1440 minutes | ANC and ALC                         |
| $m_{\text{hepatic}}$ | 1440 minutes | ALT and Total bilirubin             |
| $m_{\text{renal}}$   | 1440 minutes | creatinine                          |

Importantly, restricting the maximization to  $t \in T$  does **not** truncate the underlying data. For any  $t \in T$ , the carry-forward operator  $\mathcal{C}[\cdot](t; m)$  can still use measurements recorded before  $T$  (even at negative times) as long as they fall within the look-back window  $m$ . Conceptually, this JAMA definition is time-aligned: at each time  $t$  we sum the organ dysfunction components evaluated at that same time, then take the maximum of that sum over  $T$ . This preserves temporal alignment across organ systems, so a high score reflects concurrent multi-organ dysfunction at a single time point.

## 5 Exploratory Aggregation Schemes

There are several exploratory aggregation schemes. Daily and block-based “worst value” aggregation is common in pediatric organ dysfunction scoring (e.g., PELOD/qPELOD and pSOFA), which motivates the time-decoupled exploratory aggregation schemes below. See, for example, daily PELOD scoring using the most abnormal value each day and pSOFA evaluations that use maximum/mean daily scores or the highest score in fixed time blocks. [Leteurtre et al., 2003, Bestati et al., 2010, Matics and Sanchez-Pinto, 2017, Badke et al., 2025]

### 5.1 Organ-Level Maxima

The primary definition for PSS in (53) looks for the max of the sum of the organ dysfunction scores. This exploratory scheme looks at the sum of the max of each of the organ system dysfunction scores.

$$\text{PSS}_d^{\text{OLM}}(T; \mathbf{m}, \text{Age}) := \left[ \max_{t \in T} \text{SI}(t) \right] \left[ \sum_{x \in \Omega_d} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \right], \quad d \in \{4, 8\}. \quad (58)$$

$\text{PSS}_d^{\text{OLM}}(T; \mathbf{m}, \text{Age})$  decouples time across organ systems. It takes the maximum score for each organ system independently and then sums those maxima. This measures the overall burden of dysfunction over the interval, not necessarily dysfunction occurring at the same time. As a result,  $\text{PSS}_d^{\text{OLM}}(T; \mathbf{m}, \text{Age})$  can

be larger than the  $PSS_d(T; \mathbf{m}, \text{Age})$ , because peaks from different time points can accumulate. That is, by construction,

$$PSS_d(T; \mathbf{m}, \text{Age}) \leq PSS_d^{\text{OLM}}(T; \mathbf{m}, \text{Age}). \quad (59)$$

This follows from the max-sum inequality:

$$\max_t \sum_x f_x(t) \leq \sum_x \max_t f_x(t). \quad (60)$$

Using (58) to define sepsis is not structurally different from the primary definition (56).

$$\text{Sepsis}^{\text{OLM}}(T; \mathbf{m}, \text{Age}) := \mathbb{I} \{ PSS_4^{\text{OLM}}(T; \mathbf{m}, \text{Age}) \geq \sigma \}. \quad (61)$$

Septic Shock, for organ-level maxima, is an easier calculation than the primary definition (57).

$$\text{SepticShock}^{\text{OLM}}(T; \mathbf{m}, \text{Age}) := [\text{Sepsis}^{\text{OLM}}(T; \mathbf{m}, \text{Age})] \mathbb{I} \left\{ \max_{t \in T} \mathcal{D}[\text{Card}](t; \mathbf{m}, \text{Age}) \geq \kappa \right\}. \quad (62)$$

for  $\sigma \geq \kappa > 0$ .

## 5.2 Organ-Level + Cardiovascular Component Decoupling

In this exploratory aggregation scheme we decouple all organ systems **and** the three cardiovascular components: vasoactive medications, MAP, and lactate.

$$PSS_d^{\text{CCD}}(T; \mathbf{m}, \text{Age}) := \left[ \max_{t \in T} \text{SI}(t) \right] \left[ \sum_{x \in \Omega_d \setminus \{\text{Card}\}} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) + \sum_{x \in \mathbf{C}_{\text{card}}} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \right], \quad d \in \{4, 8\}, \quad (63)$$

where

$$\mathbf{C}_{\text{card}} := \{\text{Vasos}, \text{MAP}, \text{Lactate}\}. \quad (64)$$

The construction here creates the relationship:

$$PSS_d(T; \mathbf{m}, \text{Age}) \leq PSS_d^{\text{OLM}}(T; \mathbf{m}, \text{Age}) \leq PSS_d^{\text{CCD}}(T; \mathbf{m}, \text{Age}). \quad (65)$$

Sepsis and Septic Shock are then defined as:

$$\text{Sepsis}^{\text{CCD}}(T; \mathbf{m}, \text{Age}) := \mathbb{I} \{ PSS_4^{\text{CCD}}(T; \mathbf{m}, \text{Age}) \geq \sigma \}, \quad (66)$$

and

$$\text{SepticShock}^{\text{CCD}}(T; \mathbf{m}, \text{Age}) := [\text{Sepsis}^{\text{CCD}}(T; \mathbf{m}, \text{Age})] \mathbb{I} \left\{ \sum_{x \in \mathbf{C}_{\text{card}}} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \geq \kappa \right\} \quad (67)$$

for  $\sigma \geq \kappa > 0$ .

## 5.3 Full Component Decoupling

Under full component decoupling, each point-producing component is aggregated independently over  $T$  and then combined using the original organ-specific capping or maximum rules. Component scores are maximized over  $T$ ; for respiratory oxygenation, lower PFR and SFR values represent greater dysfunction and are therefore minimized.

$$\begin{aligned} \text{Resp}^{\text{FCD}}(T; m_{\text{resp}}) &:= \max_{t \in T} (\text{IMV}(t; m_{\text{resp}})) \times (\mathbb{I}\{B_3(T; m_{\text{resp}})\} + \mathbb{I}\{B_2(T; m_{\text{resp}})\}) \\ &\quad + \max_{t \in T} (\text{ORS}(t; m_{\text{resp}})) \times \mathbb{I}\{B_1(T; m_{\text{resp}})\}, \end{aligned} \quad (68)$$

where

$$\begin{aligned} B_1(T; m_{\text{resp}}) &:= \min_{t \in T} \text{PFR}(t; m_{\text{resp}}) < 400 \vee \min_{t \in T} \text{SFR}(t; m_{\text{resp}}) < 292, \\ B_2(T; m_{\text{resp}}) &:= \min_{t \in T} \text{PFR}(t; m_{\text{resp}}) < 200 \vee \min_{t \in T} \text{SFR}(t; m_{\text{resp}}) < 220, \\ B_3(T; m_{\text{resp}}) &:= \min_{t \in T} \text{PFR}(t; m_{\text{resp}}) < 100 \vee \min_{t \in T} \text{SFR}(t; m_{\text{resp}}) < 148. \end{aligned} \quad (69)$$

This scheme decouples respiratory support from oxygenation severity; the required support and oxygenation threshold need not occur at the same time within  $T$ .

The overall scores are then defined as:

$$\begin{aligned} \text{PSS}_d^{\text{FCD}}(T; \mathbf{m}, \text{Age}) &:= \\ &\left[ \max_{t \in T} \text{SI}(t) \right] \left[ \text{Resp}^{\text{FCD}}(T; m_{\text{resp}}) + \min \left\{ 2, \sum_{x \in \mathcal{C}_{\text{coag}}} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}) \right\} \right. \\ &\quad \left. + \sum_{x \in \omega_d} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \right], \quad d \in \{4, 8\}, \end{aligned} \quad (70)$$

where

$$\omega_4 := \{\text{Vasos}, \text{MAP}, \text{Lactate}, \text{Neuro}\}, \quad (71)$$

and

$$\omega_8 := \omega_4 \cup \{\text{Endo}, \text{Immu}, \text{Hepatic}, \text{Renal}\}. \quad (72)$$

Note: the contributions from neurologic, immunologic, and hepatic scores do consider more than one input from the EHR. However, because

$$\max_{t \in T} \max \{ \mathcal{D}[\text{GCS}](t; m_{\text{gcs}}), \mathcal{D}[\text{Pupils}](t; m_{\text{pupil}}) \} = \max \left\{ \max_{t \in T} \mathcal{D}[\text{GCS}](t; m_{\text{gcs}}), \max_{t \in T} \mathcal{D}[\text{Pupils}](t; m_{\text{pupil}}) \right\} \quad (73)$$

we need not explicitly decouple these component scores.

The constuction here creates the relationship:

$$\text{PSS}_d(T; \mathbf{m}, \text{Age}) \leq \text{PSS}_d^{\text{OLM}}(T; \mathbf{m}, \text{Age}) \leq \text{PSS}_d^{\text{CCD}}(T; \mathbf{m}, \text{Age}) \leq \text{PSS}_d^{\text{FCD}}(T; \mathbf{m}, \text{Age}). \quad (74)$$

Sepsis is defined as:

$$\text{Sepsis}^{\text{FCD}}(T; \mathbf{m}, \text{Age}) := \mathbb{I} \{ \text{PSS}_4^{\text{FCD}}(T; \mathbf{m}, \text{Age}) \geq \sigma \}, \quad (75)$$

for  $\sigma > 0$ .

Septic shock is defined as follows:

$$\text{SepticShock}^{\text{FCD}}(T; \mathbf{m}, \text{Age}) := [\text{Sepsis}^{\text{FCD}}(T; \mathbf{m}, \text{Age})] \mathbb{I} \left\{ \sum_{x \in \mathcal{C}_{\text{card}}} \max_{t \in T} \mathcal{D}[x](t; \mathbf{m}, \text{Age}) \geq \kappa \right\} \quad (76)$$

for  $\sigma \geq \kappa > 0$ .

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James E. Squires, Patrick J. McKiernan, and on behalf of the Pediatric Organ Dysfunction Information Update Mandate (PODIUM) Collaborative Squires, Robert H. Acute Liver Dysfunction Criteria in Critically Ill Children: The PODIUM Consensus Conference. *Pediatrics*, 149(Supplement\_1):S59–S65, 01 2022. ISSN 0031-4005. doi: 10.1542/peds.2021-052888I. URL <https://doi.org/10.1542/peds.2021-052888I>.

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